



Review article

Science and Metrology of defects in graphene using Raman Spectroscopy

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ABSTRACT

This article offers a historical account of the evolution of defect metrology in graphene through Raman spectroscopy over the past five decades. The application of Raman scattering to the investigation of disorder levels in graphite materials dates back to the 1970s, and substantial advancements have occurred in this field, especially following the isolation of graphene in 2006. The article starts presenting the physics related to structural defects disrupting the translational symmetry in crystalline solids, introducing a relaxation of selection rules in Raman spectroscopy that manifests as peaks induced by disorder, then it delves into significant milestones and provides a practical summary of the principal existing protocols. Furthermore, we explore the contribution of tip-enhanced Raman spectroscopy to gaining deeper insights into fundamental aspects of defects in graphene materials, owing to its capacity for spectroscopic measurements with high spatial resolution. In conclusion, we outline prospects for the further utilization of this innovative technique in enhancing both the science and metrology of defects in graphene and its applications in other two-dimensional systems.

1. Introduction

Controlling the properties of graphene by introducing structural modifications, such as ion implantation [1], patterning [2–4], pattern growth [5] or substitutional doping [6–8] is crucial for device fabrication and information processing. Fig. 1 schematically shows the concept that, on a two-dimensional system, zero-dimensional (point) and one-dimensional (line) defects can be introduced to engineer the physical properties of graphite. Point defects can be vacancies [9], substitutional atoms [10], 7–5 defects [11], among others [12]. When line defects fully surround a graphene area, it defines a graphene nanoflake, and the graphene border itself can be a type of line defect [13–16].

From a bottom-up perspective, the great challenge for graphene related nanotechnology is the development of nanodevices and, as important as the device-building method itself, is the development of accurate and precise characterization tools and protocols [17,18]. Raman spectroscopy, both in the micro- and nano-scale resolution, is outstanding for nanoscience and nanotechnology because it uses a non-invasive probe (visible light), it can easily reach 1 μ eV spectral resolution and it is a comparatively inexpensive and easy-to-perform experimental tool [19–21]. Graphene stands as the prototype material [22] where meaningful analyses, such as the relations between Raman

band frequencies or between peak integrated intensities and widths have been shown to accurately measure the number of AB-stacked layers [23,24], the amount of strain and doping [25,26] and the amount of point-like and line-like defects [27] in this perfectly surface-like two-dimensional system.

For instance, the colors in Fig. 2 are related to the graphene spectral characteristics, more specifically, to the intensity of the so-called D, G and 2D Raman spectral features that were imprinted in a graphene sheet using a He-focused ion beam [28]. Relating the intensity of these three features with different degrees of colors enables us to deliver the visual information of the world map.

On the other hand, from a top-down perspective, a significant challenge lies in achieving scalable production of high-quality materials [29]. This scalability can only be accomplished by optimizing critical parameters such as homogeneity, chemical composition, and degree of disorder [30], ensuring that the produced material meets the expectations associated with the envisioned applications. Therefore, a robust and comprehensive statistical analysis is of utmost importance in order to effectively control the production parameters and attain the desired material properties [31]. Once again, the application of

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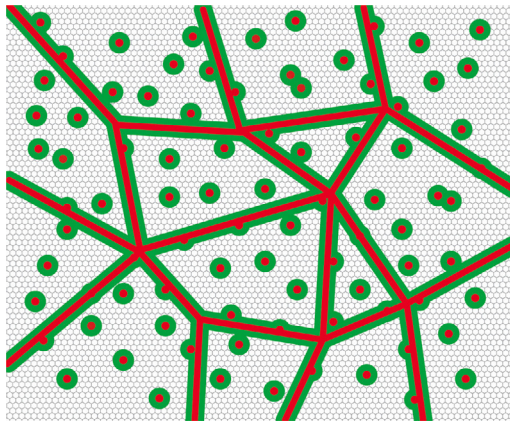


Fig. 1. Illustration of a fictional sample containing both point defects and line defects. Source: Adapted from [27].

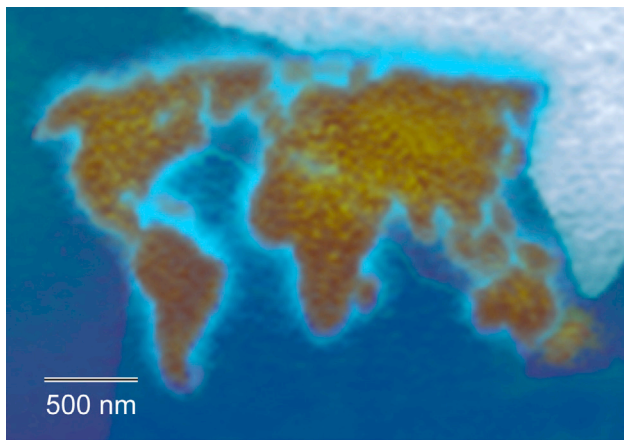


Fig. 2. World map drawn on a graphene sheet using a He-focused ion beam, and imaged using tip-enhanced Raman spectroscopy. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.) Source: Adapted from [28].

micro- and nano-Raman spectroscopy emerges as a highly valuable and indispensable tool for this purpose.

Therefore, in this work we focus on the science and metrological aspects of defect characterization in graphene using Raman spectroscopy, ending with a perspective on how this field can still develop to be increasingly used for further scientific and technological advances, and how it can be extended to other research fields.

2. Defects in graphene and related Raman spectroscopy

2.1. Defects break the translational symmetry

To comprehend the relevance of defects or structural disorder in the Raman spectrum of graphene, it is important to grasp the principle of momentum conservation in the Raman scattering process [20,32–38]. In a Raman scattering process where a phonon with wave vector $\mathbf{q}$ is generated (Stokes scattering), the wave vector of the incident photon ($\mathbf{k}_0$) must be equal to the sum of the scattered photon's wave vector ($\mathbf{k}_s$) and the wave vector of the phonon. However, the range of possible wave vectors in a solid, confined within the first Brillouin zone, is approximately on the scale of the reciprocal of the crystal's lattice parameter, which is 0.246 nm for graphene. This value is three orders of magnitude larger than the wave vector of photons in the visible range, ranging from 380 nm to 700 nm. Hence, according to momentum conservation, the wave vector of the phonon created, $\mathbf{q}$,

must be minuscule compared to the size of the first Brillouin zone, rendering it practically insignificant in relative terms. As a result, only phonons located at the center of the zone, where $\mathbf{q} \approx 0$, exhibit activity in first-order Raman spectroscopy [39]. This holds true for the G band observed in the graphene spectrum, at an approximate frequency of 1580 cm^{-1} [20], as shown in Fig. 3(a,b). The G band corresponds to a doubly degenerate stretching mode illustrated in Fig. 3(c), occurring at the convergence point of the in-plane transverse and longitudinal optical branches (iTO and iLO, respectively) at the center of the first Brillouin zone, as indicated in Fig. 3(b).

The selection rule $\mathbf{q} \approx 0$ does not apply to Raman scattering involving two phonons [39]. In this scenario, the two scattered phonons can possess opposite wave vectors, yet their sum still satisfies momentum conservation. This is particularly relevant for the 2D band, also known as G' , in the Raman spectrum of graphene [23,40]. In the case of Stokes scattering, the 2D band arises from the simultaneous creation of two almost-totally symmetric phonons that belong to the transversal optical branch, each having opposite wave vector. These wave vectors extend towards points near the corner of the first Brillouin zone, resulting in a frequency of approximately 1350 cm^{-1} . Since this process involves two phonons, the observed frequency is approximately 2700 cm^{-1} , as depicted in Fig. 3(a). The 2D band originates from a breathing-like mode, illustrated in Fig. 3(d), belonging to the in-plane transverse optical (iTO) branch near the K point, as indicated Fig. 3(b). It is generated by intervalley double- or triple-resonance processes, extensively discussed in the literature [21,32–34,36].

There is an alternative way to activate phonons outside the center of the first Brillouin zone, and it involves the breaking of translational symmetry in the crystal. To understand this, we can consider that when the crystal is no longer infinite, the concept of reciprocal space loses its validity, and the wave vector ceases to be a reliable quantum number for establishing a selection rule. However, a more concrete approach is to recognize that a localized structural defect in real space becomes delocalized in reciprocal space. Consequently, such a defect can provide momentum with a wave vector precisely opposite to that of the created phonon. As a result, even if the phonon possesses a non-zero wave vector, the scattering process can exhibit a sum of wave vectors that cancels out. This phenomenon gives rise to the emergence of the D band in the Raman spectra of graphitic materials containing defects [41], as in the case of the spectrum depicted in the top graph in Fig. 3(a). The D band manifests at an approximate frequency of 1350 cm^{-1} (for a 532 nm wavelength excitation laser). It corresponds to the same phonon created in the scattering process that gives rise to the 2D band, and it is also activated by an intervalley double-resonance mechanism [21,32–34,36]. However, instead of two phonons with opposite wave vectors, we have the creation of a single phonon, whose momentum is compensated by the scattering induced by a defect in the lattice. As we will see, this phenomenon has opened numerous possibilities to explore defects in carbon materials [19].

Additionally, there is the D' band which, similar to the D band, is activated by defects. It occurs immediately to the right of the G band at $\sim 1620 \text{ cm}^{-1}$, in the longitudinal optical (iLO) branch as indicated in Fig. 3(b). It is also generated through a double-resonance mechanism [33].

The D and D' bands measured at edges exhibit a strong dependence on the polarization of the incident light's electric field. They display maximum/minimum intensity when the electric field's polarization is parallel/perpendicular to the edge [13,14]. Another intriguing aspect of these bands is that the ratio between the intensities of the D and D' bands provides valuable information about the spatial localization nature of defects [42].

2.2. The Tuinstra and Koenig relation for average crystallite size L_a

The proposal to use Raman spectroscopy for measuring defects in graphitic materials was initially introduced in the renowned article by

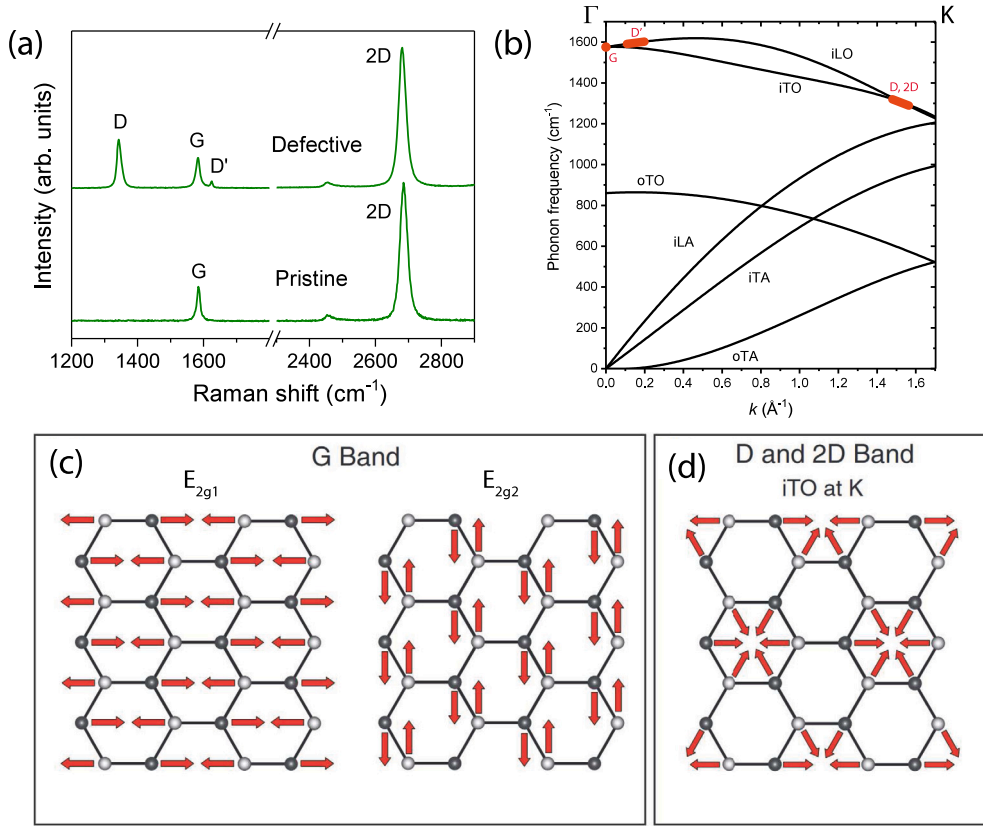


Fig. 3. Raman spectrum of graphene with (top) and without (bottom) defects. (b) Phonon dispersion of graphene along the $\Gamma - K$ direction in the first Brillouin zone [36]. Schematics of the vibration modes of (c) the two double degenerate eigenvectors related to the G band and of (d) the D band (also 2D or G') at the K point. Source: Adapted from [38].

Tuinstra and Koenig [41]. In their work, the authors not only observed the emergence of the D band in defective graphite, but also noted that the intensity of the D band increased with the degree of disorder. To further investigate, they plotted the inverse of the crystallite size (L_a) of each sample, which was previously determined using X-ray diffraction, against the ratio of intensities between the D band and the G band (Fig. 4(a)). Remarkably, they discovered a linear relationship, which they explained through a simple geometric model: while the G band's intensity is directly proportional to the area of each crystallite, the D band is specifically activated only near the edges, as depicted in Fig. 4(b), resulting in its intensity scaling in relation to the crystallite's perimeter. Consequently, the ratio between the D and G intensities is expected to be proportional to the inverse of L_a , since $I_D/I_G \propto L_a/L_a^2$. This finding gave rise to the widely recognized Tuinstra and Koenig relation [41]:

$$\frac{I_D^{(1D)}}{I_G} = \frac{\gamma^{(1D)}}{L_a}, \quad (1)$$

where the superscript (1D) indicates that the edges are one-dimensional defects.

Later on, it was observed that the proportionality constant $\gamma^{(1D)}$ in Eq. (1) depends on the wavelength of the incident laser, λ_L [43]. In a more sophisticated experimental study conducted in 2006, Cançado and colleagues empirically validated $\gamma^{(1D)} = (2.4 \times 10^{-10})\lambda_L^4$, with λ_L given in nanometers [44]. Alternatively, this relation can be expressed as $\gamma^{(1D)} = 560E_L^{-4}$, with E_L representing the excitation laser energy measured in eV units.

2.3. The amorphization trajectory by Ferrari and Robertson

In 2001, Ferrari and Robertson observed that the Tuinstra and Koenig relation fails in materials that undergo a high level of disorder,

leading to amorphization [45]. In this regime, the ratio I_D/I_G starts to decrease with increasing disorder, as illustrated in Fig. 5(a), which presents an extended version of the Tuinstra and Koenig model. The authors provided an explanation for this phenomenon by highlighting the sensitivity disparity between the D and G bands. The D band, arising from an almost fully-symmetric mode involving multiple atoms, proves to be significantly more sensitive to amorphization compared to the G band, which originates from a stretching mode involving only two neighboring atoms. Through a back-to-envelope analysis, the authors proposed the relationship $I_D/I_G \propto L_a^2$ for the highly-defective regime, as schematically depicted in Fig. 5(a).

2.4. The Lucchese's relation for average distance between point defects L_D

In 2010, Lucchese and collaborators conducted an experimental study on the amorphization trajectory applied to samples with point defects [46]. Their objective was to investigate the amorphization trajectory by introducing defects under a controlled way using low-energy Ar^+ ion bombardment on monolayer graphene. Through the use of scanning tunneling microscopy (STM), they accurately measured the average distance between point defects, referred to as L_D . Additionally, they extracted the I_D/I_G ratio from spectra obtained by employing a laser with a wavelength of 532 nm. The compelling findings are graphically presented in Fig. 5(b), illustrating the relationship between I_D/I_G and L_D . Remarkably, these results vividly support the consistency of Ferrari and Robertson's model in a qualitative picture [45]. However, it is worth noting that, for point defects occurring outside the high defective regime, that is, for $L_D \geq 4$ nm, the I_D/I_G ratio exhibits a scaling behavior with the inverse square of L_D . This is analogous to stating that the I_D/I_G ratio is proportional to the surface density of

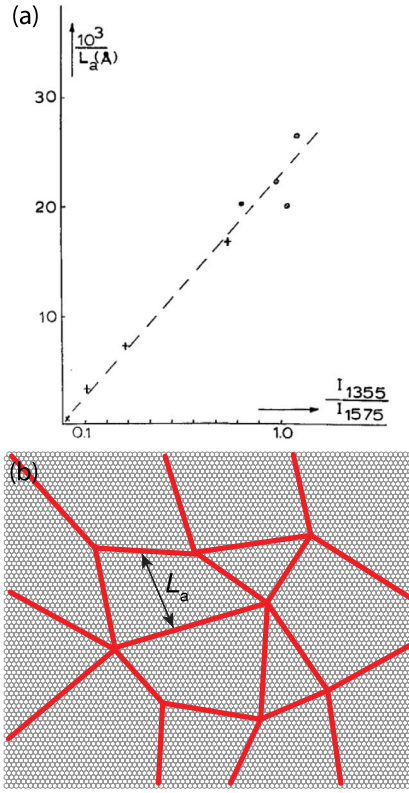


Fig. 4. (a) Plot of the inverse of the average crystallite size, L_a^{-1} , with the ratio of intensities between the D band and the G band. (b) A hexagonal network of graphene, subdivided into crystallites, with their boundaries highlighted by red lines. While the G band's intensity is directly proportional to the area of each crystallite, the D band is specifically activated only near the edges. L_a represents the average size of the crystallites. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

point defects, as systematically summarized by the following equation:

$$\frac{I_D^{(0D)}}{I_G} = \frac{\gamma^{(0D)}}{L_D^2}. \quad (2)$$

The superscript (0D) indicates that ion bombarded regions are zero-dimensional point defects. Similar to the 1D case, it was experimentally demonstrated that $\gamma^{(0D)}$ scales with the fourth-power of the wavelength of the incident laser [47]. More specifically, it was found $\gamma^{(0D)} = (1.8 \times 10^{-9})\lambda_L^4$, with λ_L given in nanometers, or, similarly, $\gamma^{(0D)} = (4.3 \times 10^3)/E_L^4$, with E_L given in eV.

The work by Lucchese and collaborators [46] made a groundbreaking contribution by proposing a phenomenological model that effectively elucidates how the Raman figure of merit I_D/I_G is related to the amorphization trajectory. The model, which lays the physical foundation for the solid line fitting the experimental data in Fig. 5(b), accurately explains the observed trend. According to it, the D band originates from two distinct regions, as illustrated in Fig. 6(a,b). The red region, defined by the radius r_S , represents the actual location of the structural defect or, in another words, the structurally damaged area. This can be observed, for instance, in areas damaged by ion impact in ion-irradiated samples, resulting in point defects. For samples consisting of crystallites, as depicted in Fig. 6(c), this region is defined by the width of the grain boundaries, ℓ_S , as defined in Fig. 6(d). The green region in Fig. 6(a-d) delineates the area in which the D band is activated by the nearby defective area, although the lattice is intact. The extent of this area is defined by the parameter ℓ_e , as illustrated in Fig. 6(b) and (d) for point and line defects, respectively. Although the

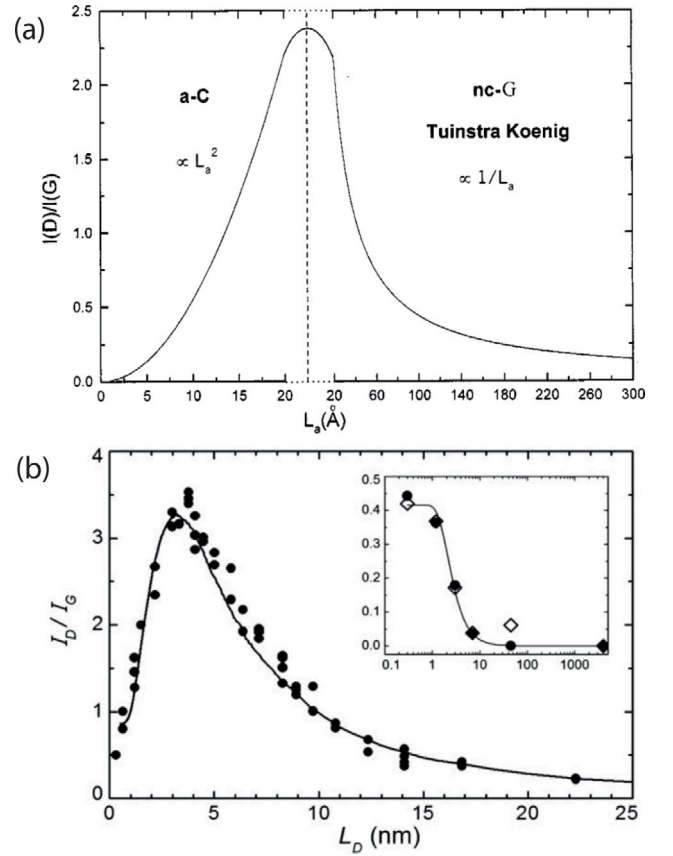


Fig. 5. (a) The amorphization trajectory proposed by Ferrari and Robertson [45]. (b) I_D/I_G as a function of the average distance between point defects, L_D , generated by Ar^+ ion bombardment on monolayer graphene [46]. The inset stands for graphite.

origin of ℓ_e is not entirely clear, many studies suggest that this length is defined by the coherence length of the electronic states involved in the scattering process [15,21,46,48].

While ℓ_S is determined by the size of the structural defect, which can be directly measured using scanning tunneling microscopy (STM) [46,48], the extension of ℓ_e is a spectroscopic property that can only be assessed through optical techniques. Its value has been indirectly determined in numerous studies [15,46,48], but a recent experiment utilizing nano-Raman in the tip-enhanced Raman spectroscopy (TERS) configuration yielded a measurement of $\ell_e = 7.8 \pm 3.2$ nm [49], as depicted in Fig. 7.

The Lucchese's model considers that the Raman response in the regions of structural damage and activation of the D band are different [46]. In other words, the ratio I_D/I_G is proportional to the red area with a strength given by a factor C_S in the structurally damaged regions defined by the radius r_S , and proportional to the green area with a strength given by a factor C_A in the activation regions defined by the length ℓ_e . Based on the arguments raised by Ferrari and Robertson [45], it is expected that the D band should be proportionally weaker in the structurally affected regions, which leads to $C_S < C_A$. By increasing the density of point defects, i.e., by decreasing the distance L_D between them, the amount of red and green area increases, resulting in an increase in I_D/I_G proportional to L_D^{-2} . This relationship works well until the distance between point defects becomes approximately twice ℓ_e . At this point, the D band is activated throughout the whole graphene structure, meaning that the green and red areas cover the entire graphene. Beyond this point, in the high-disorder regime, in which L_D becomes smaller than ℓ_e , the area of structurally damaged regions dominates, and the D band weakens relative to the G band. In

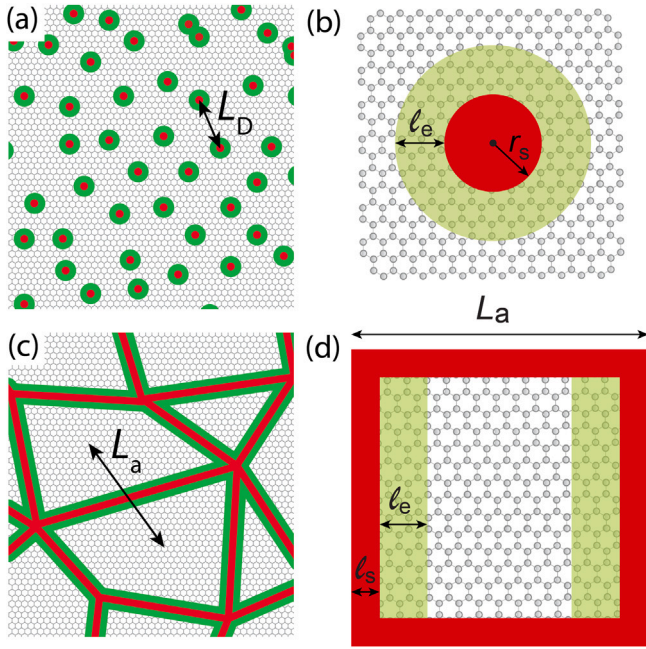


Fig. 6. Illustrations of graphene samples containing point defects (a,b) and crystallites (c,d). The red region is the structurally damaged area. The green region delineates the area in which the D band is activated by the nearby damaged area, although the graphene lattice is intact in these activated areas. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)
Source: Adapted from [27,46,48].

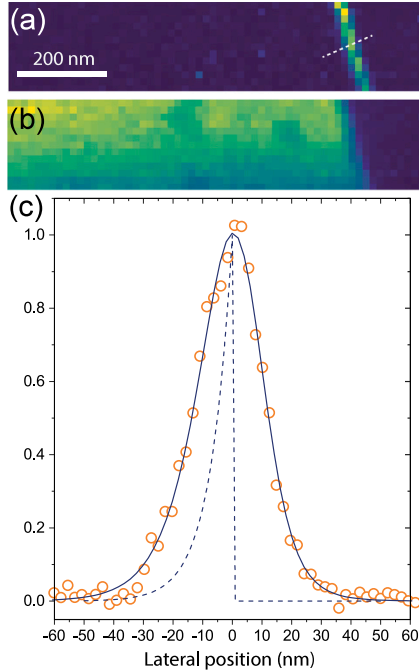


Fig. 7. (a) and (b) depict TERS intensity maps of a graphene edge derived from the same hyperspectral data set, illustrating the integrated areas of the D and 2D bands, respectively. Panel (c) exhibits the intensity profile of the TERS D band along the dashed line in panel (a). The solid dark line corresponds to the fitting of the experimental data using the theoretical framework developed in [49], with the Raman susceptibility defined by the dashed line. The fitting procedure determines $\ell_e = (7.8 \pm 3.2)$ nm.
Source: Adapted from [49].

this regime, the relationship between I_D/I_G and L_D is inverted, with the ratio becoming weaker as L_D decreases, as shown in Fig. 5.

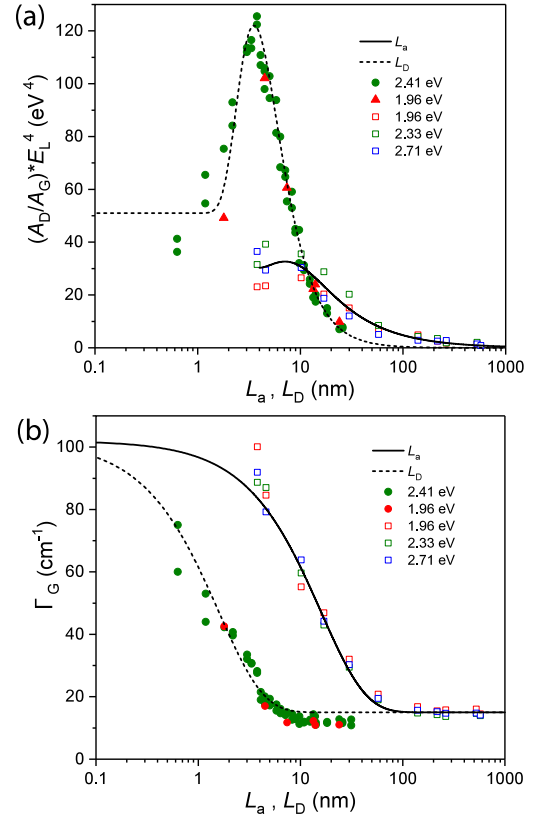


Fig. 8. (a) Experimental data plot (symbols) depicting the relationship between the A_D/A_G ratio and L_D for samples exclusively consisting of point defects (filled symbols), as well as the correlation between the A_D/A_G ratio and L_a for samples solely comprised of one-dimensional defects (empty symbols). The dashed and solid lines correspond to the fitting of the experimental data using Eqs. (3) and (4), respectively. (b) Γ_G as a function of L_D for samples exclusively consisting of point defects (filled symbols), and Γ_G as a function of L_a for samples solely comprised of one-dimensional defects (empty symbols). The dashed and solid lines correspond to the fitting of the experimental data Eq. (5). The fitting parameters are in Table 1. The colors of the symbols discriminate the energy of the incident laser used in each experiment, as indicated in the charts. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)
Source: Adapted from [27].

Based on the parameters outlined above, Lucchesse's phenomenological model resulted in the following expression [27,46]:

$$\left(\frac{A_D}{A_G}\right)^{(0D)} E_L^4 = C_S^{(0D)} \left(1 - e^{-\frac{\pi r_s^2}{L_D^2}}\right) + \frac{2\pi}{L_D^2} C_A^{(0D)} \ell_e (\ell_e + r_s) e^{-\frac{\pi r_s^2}{L_D^2}}, \quad (3)$$

It is important to highlight the change in symbol in Eq. (3), from I to A , used to denote the scattering efficiency of the D and G bands. While I is typically associated with the peak height of the band, A represents the integrated area. As discussed in [47], the integrated area offers a more comprehensive information about the scattering process, making it a more appropriate choice for this type of analysis.

2.5. The Soares' relation for average size of crystallites L_a

In 2015, Soares et al. proposed a similar model that establishes a connection between the A_D/A_G ratio and the crystallite size, L_a , along the entire amorphization trajectory, giving rise to the following relationship [27,48]:

$$\left(\frac{A_D}{A_G}\right)^{(1D)} E_L^4 = \frac{1}{L_a^2} \left[4C_S^{(1D)} \ell_s (L_a - \ell_s) + 2C_A^{(1D)} \ell_e (L_a - 2\ell_s) \left(1 - e^{-\frac{L_a - 2\ell_s}{\ell_e}}\right) \right]. \quad (4)$$

Table 1

Numerical values for the parameters found after fitting experimental data shown in Fig. 8 with Eqs. (4), (3), and (5) [27].

$C_F = 87 \text{ cm}^{-1}$
$\Gamma_G(\infty) = 15 \text{ cm}^{-1}$
$\gamma_{0D} = 1.6 \text{ nm}$
$\gamma_{1D} = 16 \text{ nm}$
$C_S^{1D} = 30.3 \text{ eV}^4$
$C_A^{1D} = 30.4 \text{ eV}^4$
$\ell_e = 4.1 \text{ nm}$
$\ell_s = 2 \text{ nm}$
$C_S^{0D} = 51 \text{ eV}^4$
$C_A^{0D} = 26.5 \text{ eV}^4$
$r_s = 2.2 \text{ nm}$

Fig. 8(a) shows the experimental data plot (symbols) depicting the relationship between the A_D/A_G ratio and L_D for samples exclusively consisting of point defects (filled symbols), as well as correlation between the A_D/A_G ratio and L_a for samples solely comprised of one-dimensional defects (empty symbols). The dashed and solid lines correspond to the fitting of the experimental data using Eqs. (3) and (4), respectively. Table 1 presents the obtained parameters extracted from the fitting processes [27].

2.6. Another important parameter: the G band linewidth

Fig. 9(a) illustrates the progression of Raman spectra acquired from samples exhibiting different L_a values (left) and L_D (right) [27]. These graphs serve as a valuable visual aid in comprehending the variations in the A_D/A_G ratio as disorder levels change. As can be seen, A_D/A_G increases as L_a and L_D decrease, following the Tuinstra and Koenig relationship [41], reaching a maximum value at $L_a \approx 10 \text{ nm}$, or $L_D \approx 5 \text{ nm}$, and then decreases for smaller values, following qualitatively the amorphization trajectory proposed by Ferrari and Robertson [45]. We can observe that in addition to affecting the intensities of the D and G bands, the reduction in L_a and L_D also leads to peak broadening. This phenomenon is attributed to the confinement of phonons, which occurs when the average size of the crystallite L_a , or the average distance between point defects, L_D , become smaller than the coherence length ℓ_{ph} of the phonons involved in the scattering process [27,48,50]. The underlying physics can be explained using Heisenberg's uncertainty principle. In terms of the energy/time uncertainty, the confined phonon has a shorter lifetime, resulting in an increased energy uncertainty, consequently causing band broadening. Similarly, in terms of the momentum/position uncertainty, spatial confinement induces an increase in momentum uncertainty, relaxing the wave vector and leading to broadening of the bands.

Fig. 8(b) illustrates the plot of the width of the G band, Γ_G , as a function of L_a (empty symbols) and L_D (filled symbols). It is evident that broadening occurs for L_a values below 30 nm, suggesting that this approximate value represents the coherence length of the optical phonons. This finding aligns well with results obtained from TERS measurements in graphene [51,52], where the coherence length can be determined as the probe approaches the sample. This phenomenon will be discussed in detail in Section 4.1.

Based on a phenomenological model, Soares and colleagues proposed an exponential decay of the width of the G band with increasing L_a or L_D [27,48]. Building upon this model, the experimental results presented in Fig. 8(b) were successfully fitted (dashed line for L_D and solid line for L_a) according to the following relation [27,48]

$$\Gamma_G = \Gamma_G(\infty) + C_F e^{-L/\gamma}, \quad (5)$$

where L refers to the values of L_a or L_D for samples with 1D or 0D defects, respectively. $\Gamma_G(\infty)$ is the bandwidth of the G band for pristine graphene, fixed as 15 cm^{-1} [27]. Experimental data obtained using three different laser lines, shown in Fig. 8(b), demonstrated that

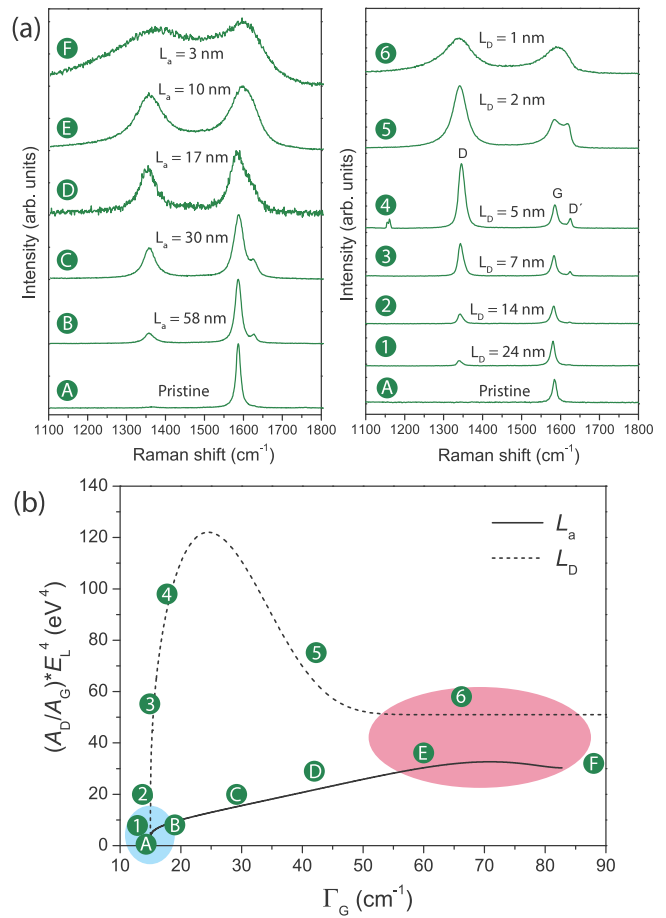


Fig. 9. (a) Progression of Raman spectra acquired from samples exhibiting different L_a (left) and L_D (right) values. (b) Relationship between A_D/A_G and Γ_G , referred here as the whale-plot. The dashed line represents the plot of $A_D/A_G \cdot E_L^4$ obtained from Eq. (3) against Γ_G obtained from Eq. (5) for the corresponding L_D values, considering the parameters listed in Table 1. The solid line is similar, but it is based on Eq. (4) and takes into account the corresponding values of L_a . The lettered and numbered circles indicate the corresponding spectra shown in (a). The blue shaded region corresponds to pristine graphene, while the red shaded region corresponds to amorphous carbon. The area bounded by the two curves represents samples that exhibit both point and line defects. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Source: Adapted from [27].

this relationship is not significantly dependent on the laser wavelength used, at least in the optical range [27,48]. The obtained decay constant values in this study were $\gamma_{0D} = 1.6 \text{ nm}$ for 0D defects and $\gamma_{1D} = 16 \text{ nm}$ for 1D defects [27]. The value of C_F obtained was 87 cm^{-1} , as indicated in Table 1 [27].

2.7. Disentangling point and line defects — the whale-plot

From Fig. 8(a), it can be noted that the same value of A_D/A_G can be observed in two different samples: one with point defects (referred to as 0D, zero-dimensional defects) and the other with edge defects (referred to as 1D, one-dimensional defects). Similarly, the width of the G band can yield the same value for both types of defects, as evidenced in Fig. 8(b). This inherent ambiguity was effectively addressed in 2017 by Cançado and collaborators [27], who systematically introduced the width of the G band as a parameter capable of distinguishing Raman spectra obtained from systems including both point defects and edge defects. With two spectral observables (A_D/A_G , Γ_G), it is possible to disentangle the contribution from the two geometrically different defects measured by (L_a , L_D). Shortly, the authors found that,

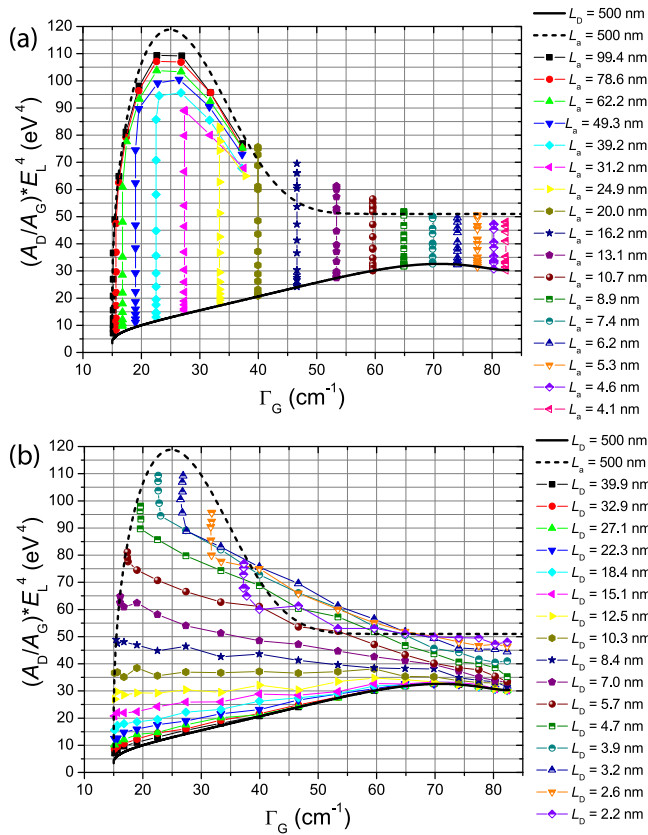


Fig. 10. Values of L_a and L_D obtained through numerical simulation. In (a), L_D varies for several fixed values of L_a . In (b) the contrary (see legends). Source: Adapted from [27].

although multiple values of A_D/A_G and Γ_G coincide between samples containing solely point defects and those with edge defects, none of the analyzed samples exhibit the same combination of A_D/A_G and Γ_G . This is illustrated in Fig. 9(b), which demonstrates the relationship between A_D/A_G and Γ_G . We denote this plot as whale-plot, since it resembles the silhouette of the animal. The dashed line represents the plot of $A_D/A_G * E_L^4$ obtained from Eq. (3) as a function of Γ_G obtained from Eq. (5) for the corresponding L_D values, considering the parameters listed in Table 1. The solid line is similar, but it is based on Eq. (4) and it takes into account the corresponding values of L_a [27].

In Fig. 9(b), the lettered and numbered circles indicate the corresponding spectra shown in Fig. 9(a). The blue shaded region corresponds to pristine graphene/graphite, while the red shaded region corresponds to amorphous carbon. The area bounded by the two curves represents samples that exhibit both point and line defects. Each point within this area corresponds to a specific (L_a, L_D) pair. Fig. 1 provides a schematic illustration of this type of sample. Through numerical simulations [27], various values of the (L_a, L_D) pairs were estimated, and the results are presented in Fig. 10(a) for fixed values of L_a and in Fig. 10(b) for fixed values of L_D (see figure legends).

Several noteworthy observations regarding Fig. 10(a) and (b). As previously mentioned, the points were determined through numerical simulation, and the specific details are elucidated in Ref. [27]. These points function as crucial benchmarks for estimating the values of L_D and L_a , respectively, based on the A_D/A_G and Γ_G values extracted from the Raman spectrum. Eqs. (3) and (4) represent analytical approximations with parameters finely tuned for samples exclusively with point and line defects, respectively. However, they fall short in reproducing the values of L_D and L_a obtained through simulation within the confines of the whale-plot. In other words, they do not adequately capture scenarios involving samples containing both types of defects.

It is crucial to emphasize that the protocol yielding the whale plot does not consider the number of layers. It was developed exclusively based on models and experimental data derived from single-layer graphene. In 2010, Martins-Ferreira and collaborators presented an exploratory study delving into the impact of layer count on the A_D/A_G ratio in samples featuring point defects [53]. In this context, the authors elucidated the evolution of Raman features induced by defects in graphene systems subjected to varying doses of argon-ion sputtering across monolayer, bilayer, trilayer graphene and graphite. The result for graphite ($N \rightarrow \infty$) is shown in the inset to Fig. 5(b). Nevertheless, this field remains open, and more comprehensive investigations are imperative to establish quantitative protocols that explicitly account for the number of layers when quantifying defect density through Raman spectroscopy.

2.8. On the nature of defects

The protocols presented here address structural defects without distinguishing the nature of the specific defect. However, these defects can originate from various sources, either purely structural, such as vacancies [46,53,54] and grain boundaries [48,55], or from a chemical perspective, like oxidation, hydrogenation, or fluorination [56], typically associated with the formation of sp^3 bonds [19]. In 2011, Eckmann and collaborators shed light on this topic in a study utilizing Raman spectroscopy to delve into the nature of these defects [42]. The authors observed that the ratio between the intensities of the D and D' bands can yield valuable insights into the type of defect. Specifically, it was noted that while the ratio of their integrated intensities, $A_D/A_{D'}$, remains unaffected by the defect density, it does vary based on the physicochemical origin of the defect. More precisely, experimental results revealed $A_D/A_{D'}$ ratios of approximately 13, 7, and 3.5 for sp^3 defects, vacancies, and grain boundaries, respectively.

In a theoretical work dating back to 2011 [36], authored by Venezuela and colleagues, a compelling explanation was offered for the impact of defect types on the $A_D/A_{D'}$ ratio. The authors postulate that charged defects exhibit Coulomb potentials with a gradual decay as measured from their center, in contrast to purely structural defects where the potential drops abruptly. Consequently, defects containing charged impurities are expected to be less spatially confined compared to purely structural defects like vacancies. Examining reciprocal space, calculations indicate that the typical wave vector associated with defects caused by charged impurities manifests near the center of the first Brillouin zone, leading to a more efficient activation of the double-resonance process responsible for the D'. Conversely, purely structural defects, being highly localized in direct space, are linked to wave vectors extending to the corner of the first Brillouin zone, thereby favoring the activation of the double-resonance process giving rise to the D band. This prompts the authors to predict a higher $A_D/A_{D'}$ ratio for structural defects than for those induced by the presence of charged impurities. While this aspect has been extensively explored in the literature, there remains a crucial need for systematic studies to establish protocols enabling the utilization of Raman spectroscopy for a quantitative investigation into the nature of defects.

3. On the relationship between amount of defects and number of layers in mass-produced graphene

The whale-plot can in principle be used to quantify point and line defect distances in a range of samples. Below, we show the applicability to mass-produced few-layer graphene to illustrate its potential.

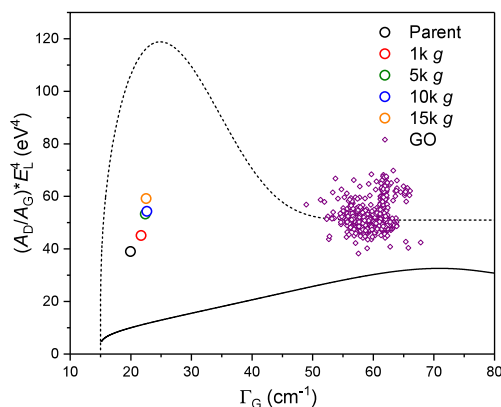


Fig. 11. Empty circles indicate the $(A_D/A_G, \Gamma_G)$ pair for a group of samples obtained through centrifugation-based stratification of mass-produced graphene using the liquid-phase exfoliation (LPE) method. Starting from the parent sample, the centrifugal force value is indicated in the legend for each stratified sample. The symbol k g means the centrifugal acceleration in units of $10^3 \times 9.8 \text{ m/s}^2$. The empty diamonds represent data extracted from a graphene-oxide sample.

3.1. Micro-Raman spectroscopy measurements

Fig. 11 locates, in the whale-plot, the values of the A_D/A_G ratio and Γ_G (empty circles) for a group of samples obtained through centrifugation-based size selection of mass-produced graphene using the liquid-phase exfoliation (LPE) method [30]. The process begins with the parent sample (also sometimes denoted stock), which is subjected to centrifugation at different centrifugal accelerations. This removes larger/thicker nanosheets as sediment with smaller/thinner ones collected as supernatant. The size-cut-off is determined by the centrifugal acceleration, which is indicated in the legend for each size-selected sample. The symbols (empty circles) actually represent the average values of the pair $(A_D/A_G, \Gamma_G)$ obtained from hundreds of individually measured flakes in each sample, performing hyperspectral mapping. It is well-known that increasing centrifugal force selects graphene flakes with a decreasing average number of layers [30]. This data demonstrates that thinner flakes also exhibit a higher degree of disorder, indicating that there is a trade-off between the level of exfoliation and the amount of defects introduced in the nanoflakes. This can be either line defects due to the finite lateral flake size, which also decreases in centrifugation, or point defects. However, the approximately vertical trajectory on the graph indicates that exfoliation introduces mostly point defects in the samples. These are valuable information for the development of the liquid-phase exfoliation methodology.

The empty diamonds in **Fig. 11** represent data extracted from a graphene-oxide sample. In this case, it is clearly visible that the samples exhibit a significant degree of amorphization caused by the oxidation process. Notably the data falls in the dashed line of the whale-plot, consistent with amorphization through addition of point-like defects related to C-O sites. This is in contrast to LPE nanoflakes which have a significantly lower defect density, making it possible to use the whale plot to distinguish between point and line defects.

3.2. Nano-Raman spectroscopy measurements

Structural defects typically manifest on the nanoscale, posing a challenge for analysis due to the limitation of low spatial resolution imposed by light diffraction. This constraint hinders the detailed examination of objects with dimensions smaller than half the wavelength of visible light, i.e., objects typically smaller than 200 nm [57]. Near-field optical spectroscopy employs a nanoscale antenna positioned just a few nanometers from the system under investigation. This nanoantenna, often a metallic nanostructure with unique plasmonic properties,

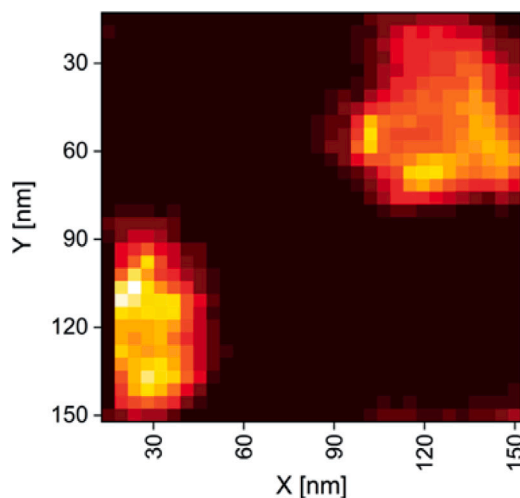


Fig. 12. Nano-Raman (TERS) map of two individual graphene nanoflakes. The map comprises 128×128 pixels, covering a $150 \times 150 \text{ nm}^2$ area. Each pixel corresponds to a spectrum (hyperspectral imaging) and, in this specific diagram, the color intensity represents the intensity of the D band, centered at 1350 cm^{-1} . The bottom left flake is designated as flake #1, and the top right flake as flake #2. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

locally illuminates and collects the scattered signal, facilitating spectral studies with nanoscale resolution [58]. The adaptation of this technique to nano-Raman spectroscopy is referred to as tip-enhanced Raman spectroscopy (TERS) [59]. Not surprisingly, graphene stands as a prototype material and it has been largely studied with TERS [16,51,59–68].

Fig. 12 shows a TERS map depicting two isolated graphene nanoflakes, extracted from the LPE sample centrifuged at 15k g. The map comprises a grid of 32×32 pixels, covering an area of $150 \times 150 \text{ nm}^2$. Each pixel corresponds to a spectrum obtained through hyperspectral imaging and, in this specific diagram, the color intensity represents the strength of the D band. We have designated the bottom left flake as flake #1 and the top right flake as flake #2. To the defect density, we conducted a systematic analysis of the A_D/A_G ratio as a function of Γ_G , and the results are presented in **Fig. 13**. As depicted in the whale-plot, the data obtained from the inner regions of flake #1 and flake #2 (represented by solid squares) exhibit distinct characteristics in terms of line and point defects. Notably, flake #1 displays a higher defect density, as evidenced by higher values of both A_D/A_G and Γ_G . In contrast, the data extracted from the edges of both nanoflakes (represented by empty triangles) show relatively low values of A_D/A_G and wider values of Γ_G , following the distinctive trajectory of line-defects spectra (solid line). This analysis distinctly reveals the non-uniform nature of the nanoflakes, highlighting that conventional Raman spectroscopy measurements yield only an average. Conversely, the remarkable spatial resolution attained with TERS empowers us with detailed insights. These insights are pivotal for refining production processes and comprehending the system comprehensively, as elucidated in the forthcoming sections.

3.3. Number of layers using principal component analysis

Beyond the defect analysis outlined earlier, Raman spectra of graphene and few-layer graphene are frequently employed to evaluate the number of layers, primarily relying on the line shape of the 2D band. This effect is illustrated in **Fig. 14**, which showcases the 2D band extracted from standard samples with different numbers of layers, previously measured using atomic force microscopy [24]. The figure demonstrates a clear evolution in the shape of the 2D band as the number of layers increases, with the monolayer exhibiting a highly

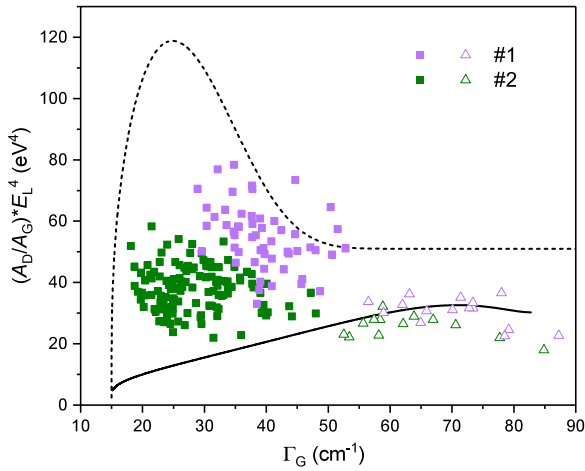


Fig. 13. A_D/A_G as a function of Γ_G in the whale-plot, depicting data extracted from nano-flakes #1 and #2 in Fig. 12. Solid squares and empty triangles represent data points from the internal regions and edges of the flakes, respectively.

symmetric Lorentzian shape. This change in shape is attributed to variations in the vibrational and electronic dispersions resulting from the increasing number of layers [23,40].

While the qualitative understanding of the effect based on the double-resonance model is well-established [21,32–34,36], performing precise theoretical calculations to reconstruct the shape of the 2D band for two or more layers is complex [69] and generate results that are not experimentally accurate [70]. Recent studies have proposed a more practical approach to determine the number of layers using a parameterized principal component analysis (PCA) method [24,71]. In essence, the number of layers can be classified by comparing spectra from unknown samples with spectra obtained from standard samples. This systematic procedure has been demonstrated by applying the PCA model to mass-produced graphene samples using liquid-phase exfoliation, and other machine learning-based methods can be employed in a similar fashion.

We conducted the PCA analysis based on the shape of the 2D band measured on the nanoflakes shown in Fig. 12, following the 2D protocol described in Ref. [24]. The PCA algorithm initiates by constructing an $n \times m$ matrix, wherein spectral intensities are recorded as elements. The rows represent sample spectra, while the columns correspond to wavenumbers. A transformation is applied to the coordinate space, defining a new basis comprising principal components (PC_i , with $i = 1, 2, \dots, m$). Subsequently, the experimental data is reformulated on this new basis, representing each data point as a linear combination of the principal components. The resulting PCs are expressed in spectral forms that encapsulate specific spectral features, enhancing subtle variations associated with distinct properties of the sample. This renders them more discerning and sensitive than the original spectra. The results are presented in Fig. 15, which depicts the plot of the first and second principal components (PC_1, PC_2) extracted from each spectrum. The number of layers, denoted as N , is determined by calculating the probability of a pixel belonging to a specific class represented by spectra obtained from standard samples using the PCA-parameterized method [24]. As demonstrated in the graph, the majority of spectra extracted from flake #1 exhibit similarity to spectra obtained from standard samples with 2 layers, whereas the spectra extracted from flake #2 resemble spectra measured from standard samples with 5 or 6 layers.

Fig. 16 showcases two graphs that effectively summarize the two key findings of the combined analysis of defects and number of layers. In Fig. 16(a), the intensity map of the A_D/A_G ratio is displayed, clearly demonstrating that flake #1 exhibits a higher defect density compared to flake #2. In Fig. 16(b), the color scale is associated with the average

number of layers, denoted as N , derived from the parameterized PCA analysis (Fig. 15). This visualization distinctly shows that flake #1 consists, on average, of 2 layers, whereas flake #2 consists of 5 or 6 layers. These results suggest a correlation between thinner flakes (with fewer layers) and increased amount of defects. This association, while intuitive, has not been experimentally verified at the level of isolated flakes before. Notably, it should be emphasized that the success of this analysis stems from the high spatial resolution of TERS (approximately 10 nm). It is worth noting that the original nano-Raman image used for this analysis spans a lateral extent of 150 nm (Fig. 12). In contrast, a high-performance micro-Raman optical microscope possesses a focus diameter on the order of 250 nm if equipped with a high-NA of 1.4 or higher.

3.4. Relation between number of layers and amount of defects

Based on the previous findings, an unprecedented statistical analysis was conducted to examine the correlation between point-defect density and number of layers for the same set of LPE samples analyzed in Fig. 11 (parent, 1k g, 5k g, 10k g, and 15k g). To establish a direct correlation between defect density (n_D) and AFM-height (h), we determined the number of layers for each measured flake using the G band intensity, through a method denoted as G-protocol, described in Ref. [24]. The outcomes are illustrated through the n_D versus h graphs depicted in Fig. 17(a–e). The defect density was evaluated utilizing the A_D/A_G ratio, as outlined in the procedure described in Section 2.7. Each data point on these graphs is the result of the arithmetic average value obtained from hundreds of flakes with the same number of layers in each sample. The graphs clearly illustrate the dependence of n_D on h , revealing a systematic decrease in n_D as h increases. This means thicker flakes are less defective on the basal plane. Moreover, the maximum value of n_D , typically occurring at $h = 0.334$ nm (which correspond to the thickness of a monolayer), becomes progressively larger in samples isolated at higher centrifugal acceleration. It is noteworthy that when employing higher centrifugation speeds, flakes exhibit a smaller lateral size. This suggests that nanoflakes with a smaller lateral dimension tend to contain more point defects compared to their larger lateral counterparts of the same thickness. The graphs are displayed on a logarithmic scale, indicating a power-law relationship between n_D and h in the form:

$$n_D = C h^{-k}. \quad (6)$$

Fig. 17(f) displays the values of k extracted from the linear fits presented in Fig. 17(a–e) (dashed straight lines), based on Eq. (6). These values are plotted as a function of the mass proportion, $\rho(\%)$, which quantifies the volumetric percentage of mono- and few-layer graphene within the sample, specifically representing the mass proportion of flakes containing no more than 10 layers. It was determined using atomic force microscopy, relying on a rigorous statistical approach that analyzed around 10^4 flakes from each sample, adhering to the protocol outlined in Ref. [72]. The graphic in Fig. 17(f) clearly illustrates that k increases as the volumetric proportion of graphene rises.

The occurrence of the power-law relationship (6) can be rationalized by a series of possible assumptions. Firstly, it is assumed that defects exclusively form post-cleavage, gradually over time. Consequently, flakes with longer lifetimes (in the state of a certain thickness), τ , are expected to exhibit a greater number of defects. For instance, if the probability of fragmentation ($1/\tau$) is directly proportional to a cross-sectional area predominantly governed by h , we have $n_D \propto h^{-1}$. Secondly, as the shearing process unfolds, the predominant mechanism driving defect formation is deemed to be the fraction of bending deformation energy within the initial layer. Given that this fraction diminishes with $1/h$, once again we derive $n_D \propto h^{-1}$. Lastly, let us consider a scenario where the dominant defect-forming mechanism during the shearing process involves a critical strain induced by a characteristic stress. Given that the Young's modulus correlates with h ,

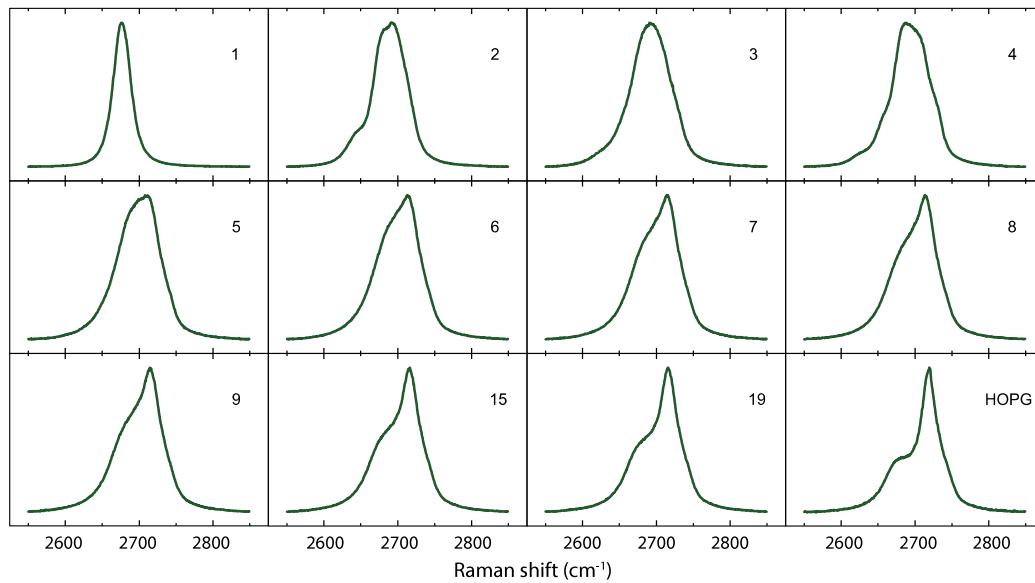


Fig. 14. 2D band spectra from graphene samples with varying numbers of AB-stacked layers N , indicated by the number in the top-right corner of each panel. HOPG stands for highly oriented pyrolytic graphite. Data was obtained using a 532 nm excitation laser line.
Source: Extracted from Ref. [24].

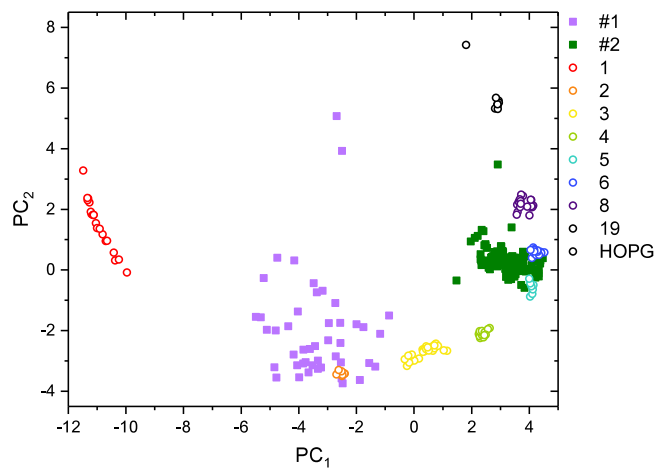


Fig. 15. Plot of the first and second principal components (PC_1, PC_2) for the measured spectra of flakes #1 and #2 of Fig. 12 (represented by purple and green squares, respectively). The data represented by filled circles are extracted from spectra obtained from standard samples with layer numbers previously established using AFM (PCA-parameterized method [24]). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

we once more arrive at $n_D \propto h^{-1}$. When combined, these three proposed mechanisms might collectively yield a relationship of n_D proportional to h^{-3} . As shown in Fig. 17(f), this closely aligns with the findings observed in the sample with the highest graphene content, specifically the one subjected to centrifugation at 15k g. It is worth noting that these principles are also applicable to other liquid-based exfoliation processes, such as sonication.

The results presented in this section indicate that the TERS analysis shown in Figs. 12 and 16 reflects a universal characteristic of liquid-exfoliated graphene: as nanoflakes decrease in thickness, they demonstrate an increased defect density. This correlation corresponds to the introduction of defects exclusively within the exposed graphene layers during the exfoliation procedure. Such information and protocol for the analysis is of immense importance for adjusting and optimizing production parameters to control the introduction of defects in the

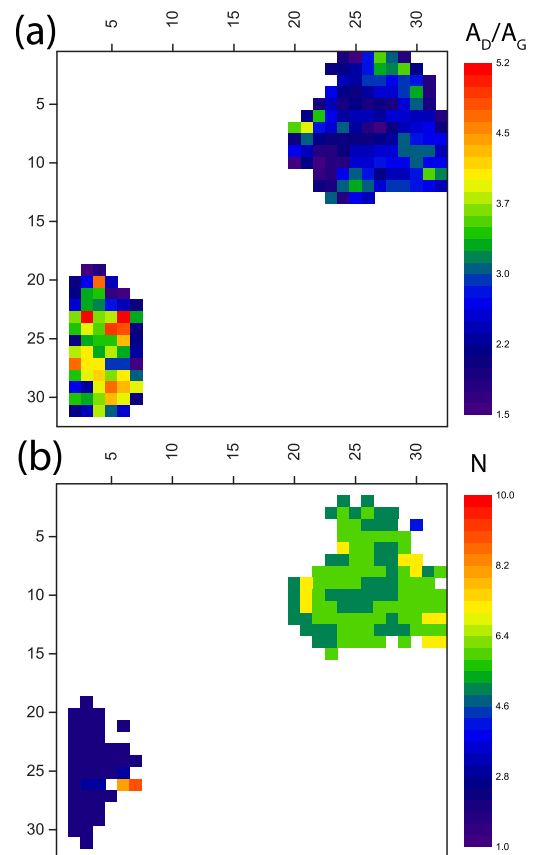


Fig. 16. (a) A_D/A_G map clearly indicating that flake #1 exhibits a higher density of defects compared to flake #2. (b) average number of layers (N) map, obtained through PCA-parameterized analysis (Fig. 15). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

material in the future, catering to specific requirements dictated by the target application.

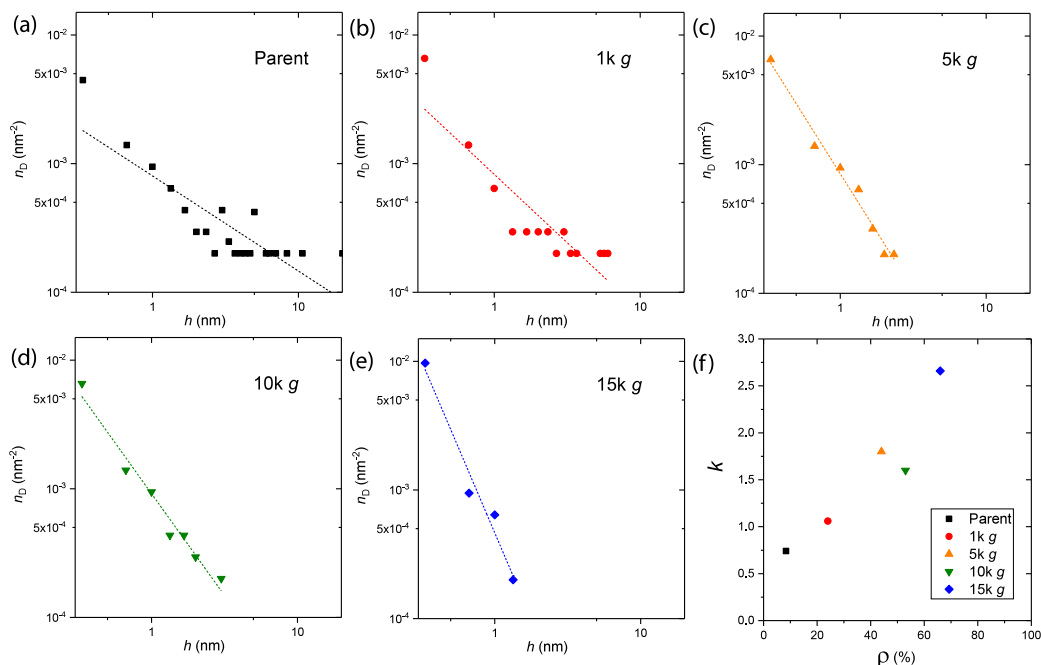


Fig. 17. (a–e) Plot of superficial defect density, n_D , versus height, h , for same set of LPE samples analyzed in Fig. 11 (parent, 1k g, 5k g, 10k g, and 15k g, respectively). The symbol kg means the centrifugal acceleration in units of $10^3 \times 9.8 \text{ m/s}^2$. Each data point on these graphs is the result of the average value obtained from hundreds of flakes with the same number of layers. (f) Values of the exponent k extracted from the linear fits presented in panels (a–e) (dashed straight lines), based on Eq. (6). These values are plotted as a function of the mass percentage, ρ (%), representing the mass proportion of flakes containing no more than 10 layers in each sample. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

4. About metrology of defects using nano-Raman spectroscopy

The scattered signal in TERS encapsulates crucial insights into the spatial coherence properties of the source. This can induce a shift in the relative intensities between Raman signals originating from vibrational modes of diverse symmetries, as exemplified by the D and G bands. In this section, we delve into the nuanced adjustment of metrological parameters for defect analysis using TERS.

4.1. Spatial coherence matters at the nano-scale

When the emitter of a wave is smaller than the wavelength, not all wavevector components of the spatial frequency spectrum propagate away from the source. Some components quickly fade, and practically vanish at distances greater than half of the wavelength, λ . The absence of such evanescent components prevents an observer located in the far-field region, that is, at distances greater than $\lambda/2$ from the source, from reconstructing its shape. This is the theoretical representation for the limited spatial resolution in optical microscopy in the range of hundreds of nanometers, and the primary motivation of most branches of near-field optical microscopy was to break this resolution limit [58].

In TERS, an optical antenna is positioned at a short enough distance from the source ($\ll \lambda/2$) to catch the evanescent components of the optical field, and thus the spatial resolution is limited by the size of the probe [73]. For similar reasons, radiation fields have a minimum correlation length of $\lambda/2$ in the far-field. Therefore, if the radiation source has spatial correlation properties whose dimensions are smaller than $\lambda/2$, such information cannot be accessed by means of conventional optical microscopy [50,74].

In a study dating back to 2014, Cançado and colleagues explained that in TERS, the probe being in nanometer proximity to the sample allows the scattered field to preserve information about the sample's coherence properties, particularly the correlation length of the source [50]. Consequently, interference effects play a crucial role in the amplified Raman signal. These interference effects can be constructive or

destructive, depending on the symmetry of the Raman mode. In the case of graphene, constructive interference occurs for the G band, while destructive interference occurs for the 2D band. As a result, as the probe approaches the sample, the enhancement of the Raman signal is more pronounced for the 2D band compared to the G band.

The significance of this signal enhancement difference increases with a larger correlation length of the source. Based on the theoretical model developed in Ref. [50] and refined in Ref. [52], this difference becomes more substantial for longer correlation lengths. Experimental results, derived from the intensity ratio of the 2D and G bands as a function of the distance between the probe and the sample indicate that the correlation length falls within the range of 30 to 40 nm for graphene [51]. While it is not entirely clear whether this correlation length is dictated by electrons or phonons involved in the scattering process [75], the dependence on phonon symmetry and the similarity to the crystallite size at which the Raman bands start to broaden lead to assumption that this correlation length is associated with the coherence length of optical phonons in graphene [50–52,76,77].

4.2. Linkage between micro- and nano-Raman spectroscopy

Fig. 18 shows the Raman spectra of (a) crystalline single-layer graphene and (b) defective graphene nanoflake. The spectra in (a,b) were measured with (red) and without (green) the presence of the TERS tip. On the evolution from micro-Raman (μR) spectroscopy to the nano-Raman (nR) regime using TERS, besides gaining resolution, other effects come into play, such as the spatial coherence explained in Section 4.1. The spacial coherence depends on the phonon symmetry and consequently the knowledge developed within the diffraction-limited far-field configuration has to be reparameterized to be utilized in the near field, because the Raman characteristics utilized to quantify defects, such as the D to G area ratio, actually change in the presence of the TERS tip [50].

Fig. 18(d) shows the A_D/A_G histogram for four different nanoflakes. The μR (green) and nR (red) data are shown on the same plot. Note that

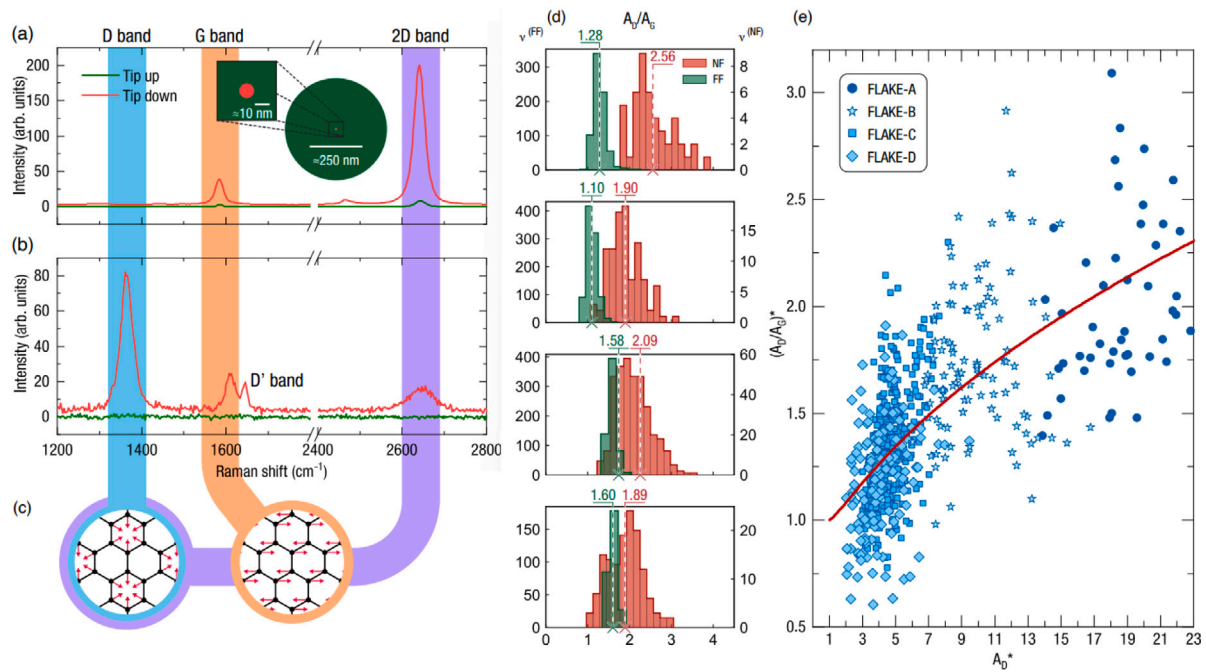


Fig. 18. (a) Raman spectra from single-layer graphene prepared by mechanical exfoliation of highly oriented pyrolytic graphite (HOPG), acquired with (nR , red) and without (μR , green) the presence of the tip. The inset shows schematically the areas from where the μR and nR spectra comes from. (b) Raman spectra from commercially available graphene nanoflake, defective as can be inferred from the magnitude of the D band, highlighted in blue, acquired with (nR , red) and without (μR , green) the TERS tip. The D' band ($\sim 1610\text{ cm}^{-1}$) is also visible. Highlighted in orange and purple is the G band and 2D band respectively. (c) G and 2D vibrational modes. (d) histogram of the (A_D/A_G) ratio for both μR (green) and nR (red) spectra acquired from different graphene nano-flakes. (e) Scatter plot of the $nR/\mu R$ normalized values $(A_D/A_G)^*$ as a function of A_D^* for the nanoflakes investigated. The solid red curve is the theoretical fit of the data points using Eq. (7). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Source: Adapted from [76].

the weighted mean values for the nR and μR distributions, represented in Fig. 18(d) by the red and green $\times$, respectively, at the x -axis, do not coincide. Flake-A shows the largest deviation, whereas Flake-D shows the smallest one. Fig. 18(e) shows a scatter plot of the $nR/\mu R$ normalized $(A_D/A_G)^*$ as a function of A_D^* for the four flakes, where the $*$ indicates the nR data was normalized by the average value of the μR data ($\bar{\mu R}$), and the area of the D band provides a metric for the spectral enhancement obtained at each nR spectrum.

The micro-to-nano spectral mismatch has been modeled using the theory of spatial coherence in near-field Raman scattering, providing a parameterization procedure to obtain consistent results. A simple formula that can be used to describe the evolution of the interference effect on $(A_D/A_G)^{nR}$ has been established [76]:

$$\left(\frac{A_D}{A_G}\right)^* = \frac{A_D^*}{1 + \alpha \left[\sqrt{1 - \beta(1 - A_D^*)} - 1 \right]}. \quad (7)$$

The solid red curve in Fig. 18(e) is the fit to the experimental data related to graphene nano-flakes through Eq. (7), giving $\alpha = (3.33 \pm 0.39)$ and $\beta = (0.58 \pm 0.09)$.

It is important to clarify that the protocol described here and the use of Eq. (7) is accurate when the defect distribution is homogeneous over the entire graphene sample. For defect inhomogeneities (see for instance [78]) the rational is not valid, and this is likely behind the data spread in Fig. 18(e), which brings $(A_D/A_G)^*$ below 1 for $A_D^* \rightarrow 1$, a behavior that cannot be described with Eq. (7). The reality is that there is still much to advance in the understanding and adjustment of parameters involved in defect metrology in graphene using TERS. This is an open field that will be explored in the coming years, especially considering the ongoing enhancement of technology in TERS instruments.

5. Conclusion and perspectives

In this paper we described the protocols for the metrology of defects in graphene. Quantifying the amount of point-like and line-like defects in single-layer graphene using micro-Raman spectroscopy is considerably well-established.

When few-layers graphene are considered, the protocols may depend on the number of layers since the sample preparation procedure may induce defects only on the layers that are at the sample surface. This is particularly important for mass-production techniques such as shear-assisted liquid phase exfoliation. Here we discussed the application of the Raman-based defect quantification on shear-exfoliated graphene containing nanosheets with varying sizes and thicknesses. However, to date the dependence of the defect characterization protocols on the number of layers is a subject that has to be studied in more detail, in particular because it might depend on the sample production technique.

We have also shown that when moving from micro-Raman to nano-Raman spectroscopy, some complexity is added to the metrology of defects, because at the nano-scale field coherence effects matters and they depend on the phonon symmetry. This is another field that requires more work to fully establish the micro-to-nano correlation. The physics behind the field coherence length, for example, is still under debate, since in Raman spectroscopy the correlation length may be dictated by the decay of phonons or of electrons [75].

The advantages of the application of TERS in graphene have also been discussed in this article. What is still missing and highly wanted is the demonstration that TERS can be utilized to identify specific defects in graphene, such as vacancies, dopants, 7–5 pairs, etc [12]. This has been done for a single doping atom in carbon nanotubes, differentiating n and p doping [79].

An even greater challenge will be to include amorphous structures with a three-dimensional (3D) character, such as diamond-like carbons

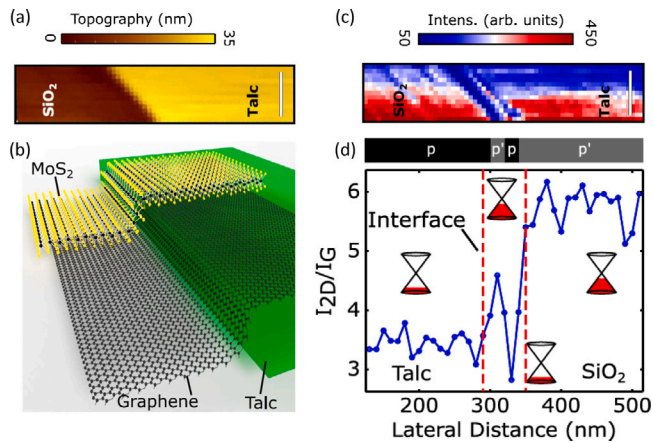


Fig. 19. (a) Atomic Force Microscopy (AFM) image depicting graphene partially resting on glass (SiO_2) and partially on talc, featuring a distinct step in the substrate transition. (b) schematics of the heterostructure. (c) Nano-Raman map of the intensity of the graphene's 2D band. The scale bar indicated 100 nm. (d) I_{2D}/I_G averaged line profile and the top panel is an illustration of the charges in the junction. Source: Adapted from [81].

(DLCs), into the nanometrology platform discussed here. Ferrary and Robertson [80] proposed a ternary phase diagram for amorphous carbons having a mixture of sp^3 , sp^2 and even sp^1 C-hybridization sites, with the possible presence of up to 60 at. % hydrogen, with DLCs defined as amorphous carbons with a significant fraction of sp^3 bonds. The degree of clustering of the sp^2 sites, which are extensively discussed in the paper, where supposed to be a fourth dimension in Ferarri and Robertson's ternary phase diagram. Addressing 3D systems with TERS is a challenge per se due to the reduced penetration depth of the near-field signal (~ 10 nm).

Finally, other areas of research can benefit from the specific science discussed in this article: the use of Raman spectroscopy to characterize other aspects of graphene-related systems, such as number of layers, stacking order, strain, doping, etc., as well as the use of Raman spectroscopy to characterize disorder in other two-dimensional systems.

5.1. Perspectives on nano-Raman to characterize other aspects of graphene-related systems

Raman spectroscopy can also be utilized to measure other effects unrelated to defects. Recently, Gadelha et al. were able to image a very fundamental aspect of semiconductor physics, which is the depletion region at the boundary between two differently doped regions on graphene [81]. They measured the TERS spectra of a homojunction formed by placing a graphene layer partly on top of a layered talc substrate, partly on top of an SiO_2 substrate (see Fig. 19). The charge oscillations observed in graphene extend over 50 nm across the junction region, and the ability to observe this comes from the implementation of nano-Raman and from the fact that the spectral characteristics of the G and 2D bands depend on doping [25,26].

Utilizing the same TERS system, the authors imaged low-angle reconstructed twisted bilayer graphene, showing that the 2D band can differentiate the AB (Bernal) spectra from areas where there is no staking order between top and bottom layers (see Fig. 20) [82]. Again, this is possible by the implementation of nano-Raman and by the fact that the 2D peak provides information about the electronic structure that changes depending on the number of layers [21]. More important, the image of localized electrons and phonons at domain walls and topological points in twisted bilayer graphene are interesting for contradicting the simplest concept that electrons and phonons are

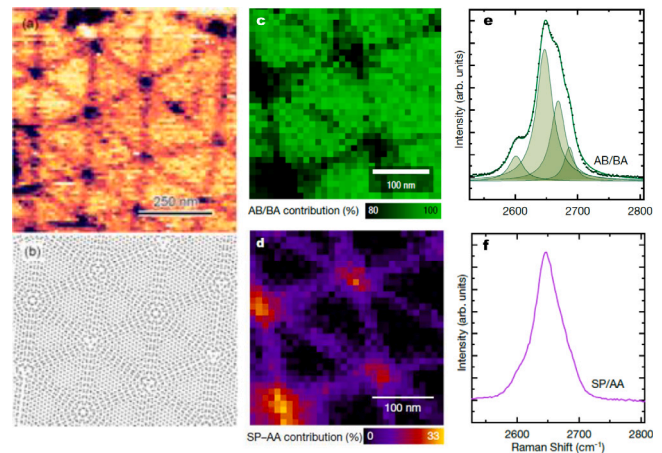


Fig. 20. (a) TERS image of domain walls (SP) and topological points (AA) in a low-angle reconstructed twisted bilayer graphene [82]. (b) Schematics of the atomic structure. (c) and (d) are the locations where the 2D band spectral shape shown in (e) and (f), respectively, are localized in the sample.

extended quasiparticles in crystals. These data also provide experimental evidence for the validity of atomistic models considering tens of thousands of atoms in the unit cell, which enables evaluation of the local density of the electronic and vibrational states of superlattices. The authors also observed changes in the phonon life-time reflecting the local influences in the electron-phonon coupling, which represent an important step towards understanding phonon-related effects at atomic and nanometric scales, such as Jahn-Teller effects and electronic Cooper pairing, and may help to improve device characterization in the context of the rapidly developing field of twistrionics.

5.2. Disorder in other 2D systems

Because 2D materials are pure surfaces, the properties of these materials and related devices are largely affected by internal and external sources of disorder [83]. Generally, graphene devices have progressed significantly with the improvements of device preparation procedures, including hBN encapsulation and the use of graphite gating. However, the remaining sources of extrinsic disorder are still to be identified, and the presence of water vapor and hydrocarbons that are not completely eliminated from interfaces during heterostructure fabrication might be identifiable with TERS. The simple deposition of graphene on a surface, if not done carefully, can generate local strain that changes the transport properties, and TERS has already been successfully utilized to identify local strains that cannot be seen with usual micro-Raman [63,84]. Intrinsic disorders in graphene, like point defects or edges, have been largely studied with Raman spectroscopy [20,27,35,38,85], and developing similar methodologies with TERS are challenging mainly because in the near-field regime, interference effects change the Raman-based figures of merit [76]. In 2D semiconductors, such as transition metal dichalcogenides (TMDs), hBN encapsulation is not as efficient as in graphene, indicating that TMD intrinsic defects are the dominant source of disorder. Mined MoS_2 , for example, can have extremely large amounts of impurities, with contractions up to 10^{14} cm^{-2} [86]. TERS is a promising technique to significantly advance the understanding and characterization of disorder in two-dimensional systems.

Furthermore, what we have learned with Raman spectroscopy applied to graphene can be generally extended to other 2D systems. For example, besides graphene, MoS_2 homojunctions formed by placing a single layer partly on top of a layered talc substrate, partly on top of an SiO_2 substrate was also made (see Fig. 19(b)). Interestingly, while the charge oscillations observed in graphene extend over 50 nm across the

junction region, in MoS₂ it is twice as large. This observation remains to be explored theoretically.

TERS measurements of correlation lengths were also performed on few-layer GaS [77]. The results showed that the correlation length decreases when decreasing the number of layers, indicating that scattering with the surface roughness plays an important role in defining the spatial coherence properties in this 2D system [77].

CRediT authorship contribution statement

Luiz G. Cançado: Conceptualization, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Supervision, Validation, Writing – original draft, Writing – review & editing. **Vítor P. Monken:** Data curation, Formal analysis, Investigation, Methodology, Writing – original draft. **João Luiz E. Campos:** Data curation, Formal analysis, Investigation, Writing – original draft. **Joyce C.C. Santos:** Formal analysis, Investigation. **Claudia Backes:** Formal analysis, Investigation, Methodology, Validation, Writing – original draft. **Hélio Chacham:** Investigation, Validation, Writing – original draft. **Bernardo R.A. Neves:** Formal analysis, Investigation, Methodology, Validation. **Ado Jorio:** Conceptualization, Formal analysis, Funding acquisition, Investigation, Project administration, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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